

Full Length Article

Similarity-guided state attention for visual reinforcement learning

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ARTICLE INFO

Keywords:

Visual reinforcement learning
 Similarity guidance module
 State attention map
 State representation
 Self-supervised learning objective

ABSTRACT

Visual reinforcement learning (VRL) aims to extract effective visual information from high-dimensional observations to optimize decision-making policies. While existing VRL methods have achieved significant progress in various control tasks through data augmentation and auxiliary tasks, agents remain susceptible to distractions from redundant information and irrelevant factors, leading to overfitting and degraded generalization in unseen environments. To address this challenge, we propose a similarity-guided state attention (SSA) for VRL. In this method, a similarity guidance module (SGM) is designed to leverage the similarity between state embeddings from the original and augmented observations to guide the encoder to focus on task-relevant regions in the original observation, thereby producing a corresponding state attention map. Meanwhile, the state attention map of augmented observation is obtained by decoding its state embedding. Furthermore, cosine similarity is introduced to measure the global similarity between state attention maps of original and augmented observations, which is incorporated into self-supervised learning objective together with binary cross-entropy loss to encourage alignment of state attention maps in representation space. The combination of SGM and cosine similarity-based alignment of state attention maps facilitates self-supervised learning to obtain more robust state representations for downstream reinforcement learning, thereby enabling the agent to learn optimal policies and improve its generalization ability in unseen environments. Experimental results on the DeepMind Control Generalization Benchmark (DMControl-GB) demonstrate that SSA achieves superior robustness and generalization performance compared with representative VRL baselines.

1. Introduction

Reinforcement learning (RL) (Xu et al., 2025; Yang et al., 2026) is one of the core areas of machine learning. Different from supervised learning, RL does not rely on ground truth, but optimizes policies based on reward signals derived from environmental feedback through the interaction and trial-and-error between agent and the environment. This characteristic enables RL to show great potential in various decision-making tasks (Hassani et al., 2026; Wang et al., 2025b). However, as task complexity and environmental diversity increase, especially in scenarios with high-dimensional visual observation inputs (such as images and videos), traditional RL methods face new challenges. Agents need to extract effective features from high-dimensional visual data, but are often interfered by redundant information and irrelevant factors.

With the rapid development of deep learning, its advantages in feature representation and generalization performance have laid the foundation for the rise of visual reinforcement learning (VRL). VRL combines deep learning, RL, and visual information processing techniques, enabling agent to better understand high-dimensional visual environments and make decisions. Specifically, deep neural networks automatically

extract visual information, transforming complex high-dimensional inputs into low-dimensional features so that agents to learn optimal policies and perform effective actions. Recent advances in visual tasks, such as visual-textual fusion for image enhancement and graph-based modeling for complex visual classification, further indicate the importance of learning robust and informative representations from high-dimensional visual data (Li et al., 2025; Wang et al., 2026). This makes VRL excellent on complex control tasks such as autonomous driving (Kiran et al., 2022; Zhang et al., 2025), robot control (Cai et al., 2023; Wang et al., 2024a), navigation tasks (Li et al., 2020; Rao et al., 2021), video games (Shao et al., 2019; Xu et al., 2022), and unmanned aerial vehicles task (Yan et al., 2022). However, how to effectively filter out redundant information and irrelevant factors, while improving the robustness and generalization ability of agent in unseen environments, remains a significant challenge in current VRL research.

In recent years, researchers have extensively explored the generalization problem in the field of VRL, with related methods broadly categorized into two main categories. The first category consists of data augmentation-based methods (Hansen et al., 2021; Laskin et al., 2020a; Raileanu et al., 2021), which draw inspiration from data

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augmentation techniques in computer vision. These methods apply transformations or perturbations to high-dimensional visual inputs to enhance data diversity, thereby improving the agent's generalization ability. By directly learning from augmented observations, they aim to increase model robustness through a richer input distribution. However, mere data augmentation cannot guarantee that the agent consistently focuses on task-relevant visual regions, limiting its ability to learn crucial feature representations and thus affecting generalization performance in unseen environments. The second category includes auxiliary task-based methods (Hansen et al., 2020; Wang et al., 2024b; Yuan et al., 2022), which often incorporate representation learning to extract high-level features from high-dimensional visual observations, and refine low-level state embeddings using contrastive loss (Laskin et al., 2020b; Zhu et al., 2023) or reconstruction loss (Ze et al., 2023; Zhang et al., 2024) to improve reinforcement learning efficiency. Typically, such methods train an encoder using data augmentation, mapping both original and augmented observations into a low-dimensional representation space, which is then fed into the policy network for decision-making. Simultaneously, they jointly optimize the model using auxiliary tasks and reinforcement learning objectives. Although these methods have achieved strong performance across various control tasks, they do not explicitly model the relationship between original and augmented observations when learning visual representations. As a result, agents may struggle to consistently focus on task-relevant regions under visual perturbations, making the learned policies susceptible to task-irrelevant factors (e.g., background, color, and position).

Some studies (Bertoin et al., 2022; Grooten et al., 2024) have incorporated attention or masking mechanisms into auxiliary tasks to improve the robustness and generalization performance by emphasizing salient regions in visual observations and suppressing task-irrelevant information. These methods have demonstrated effectiveness in guiding agents to focus on important visual features and improving generalization performance across diverse environments. However, they rely on heuristic mask generation or indirect supervision signals (e.g., reward or saliency), without explicitly modeling the intrinsic relationship between original and augmented observations. As a result, the learned attention may not accurately capture consistent task-relevant regions under visual perturbations.

To fill this research gap, we propose a similarity-guided state attention for visual reinforcement learning (SSA) in this paper, which explicitly models the similarity between original and augmented observations to guide representation learning, enabling robust and consistent identification of task-relevant representations under visual perturbations, thereby improving the agent's robustness and generalization in unseen environments. Firstly, a similarity guidance module based on visual attention mechanism is established to learn the intrinsic similarity between state embeddings original and augmented observations, thereby obtaining the attention map of original state and promoting more effective utilization of the state in downstream control tasks. Subsequently, cosine similarity is introduced to measure the global similarity between attention maps from original and augmented states, and is incorporated into self-supervised learning objective to encourage alignment of state attention maps in the representation space, thereby improving robustness under distracting visual variations. Finally, the original observation is fused with corresponding state attention map for Q-value estimation, while a consistency term (Bertoin et al., 2022) is introduced to encourage the value function to prioritize crucial pixels in the input, facilitating optimal policy learning. Furthermore, comprehensive evaluations on the DeepMind Control Generalization Benchmark (DMControl-GB) (Hansen & Wang, 2021) with two levels of visual perturbations, demonstrate that SSA consistently outperforms all VRL baselines.

To sum up, the main contributions are threefold as follows:

- 1) We propose SSA, a similarity-guided state attention for visual reinforcement learning, aiming to improve the robustness and generalization ability of agents in unseen environments.
- 2) The similarity guidance module (SGM) leverages similarity between state embeddings of original and augmented observations to guide the encoder to extract task-relevant representations, thus generating a state attention map for original observation that highlights critical regions while suppressing irrelevant distractions.
- 3) A cosine similarity is introduced to measure the global similarity between attention maps of original and augmented states, and is combined with binary cross-entropy (BCE) to form self-supervised learning objective, thereby refining state attention maps and providing more informative pixel-level support for downstream RL.

The subsequent content of this paper is organized as follows. Section 2 an overview of related work in VRL. Section 3 provides the problem definition of the proposed method, and introduces relevant preliminaries of visual attention mechanisms and self-supervised learning in VRL. Section 4 elaborates on the architecture of the similarity-guided state attention for VRL. Section 5 presents and thoroughly analyzes the experimental results. Finally, Section 6 concludes the findings and outlines directions for future research.

2. Related work

In this section, we review relevant work from two perspectives: data augmentation and auxiliary task in VRL.

2.1. Data augmentation in VRL

In VRL, data augmentation introduces perturbations to increase the diversity of state observations and is widely used to tackle generalization challenges. Different VRL algorithms are suited to different types of data augmentation strategies. Representative studies include the reinforcement learning with augmented data (RAD) proposed by Laskin et al. (2020a), which combines the soft actor-critic (SAC) (Haarnoja et al., 2018) with data augmentation techniques (e.g., random translations and random amplitude scaling) to improve the adaptability and generalization of agent. Subsequently, the stabilized Q-value estimation under augmentation (SVEA) (Hansen et al., 2021) decomposes the state-action value function into a state encoder and a Q-function. It applies data augmentation only to the current state's Q-value estimation, while the target Q-value is computed from the unaugmented next state. This helps reduce excessive regularization and improve training stability. Almuzairee et al. (2024) argue that data augmentation in SVEA is effective only for photometric variations, which limits its practical applicability. To overcome this limitation, they propose the stabilized actor-critic under data augmentation (SADA), which supports diverse augmented observations to enrich state representations. Meanwhile, SADA increases augmented inputs for both the actor and critic, effectively mitigating training instability. Further, the data-regularized Q (DrQ) (Yarats et al., 2021b) extends RAD by enabling flexible adjustment of data augmentations and incorporating two distinct mechanisms for Q-function regularization to reduce overfitting in agent. Building upon DrQ, the more efficient DrQ-v2 (Yarats et al., 2021a) is developed, offering better performance. DrQ-v2 further enhances agent's performance and sample efficiency through techniques such as bilinear interpolation, n-step returns, and exploration exploration noise. Recently, the diffusion guided adaptive augmentation (DGA2) (Lee & Hwang, 2025) leverages Stable Diffusion to generate augmented observations and introduces an adaptive domain shift strategy that dynamically adjusts augmentation strength, thereby enhancing domain diversity and improving generalization in VRL. However, these methods primarily improve generalization by increasing observation diversity, without explicitly constraining the learned representations to focus on task-relevant regions. As a result, the learned policies may still be influenced by task-irrelevant visual variations, especially when encountering unseen environments with significant environment variations. These limitations motivate the need for more effective mechanisms to guide representation learning toward task-relevant information.

2.2. Auxiliary task in VRL

Beyond data augmentation, auxiliary task-based methods have been widely explored to improve generalization in VRL by enhancing state representation learning. These methods introduce additional learning objectives to extract task-relevant information from high-dimensional visual observations and improve policy robustness. Existing auxiliary task-based VRLs can be broadly categorized into two groups: representation learning-based VRLs and attention or masking-based VRLs.

Representation learning-based VRLs aim to learn invariant and compact feature representations through auxiliary objectives such as contrastive learning, reconstruction, and policy pretraining. For example, Hansen et al. (2020) adapt policies to unseen environments by optimizing both reinforcement learning and self-supervised objectives on intermediate representations. Zhao et al. (2024) learn disentangled and interpretable representations using a β -variational autoencoder guided by reward prediction. Liang et al. (2024) further improve representation learning by modeling action sequences to guide the encoder in separating task-relevant and task-irrelevant features. In addition, Laskin et al. (2020b) propose the contrastive unsupervised representation for RL (CURL), which utilizes contrastive learning to extract features from observational inputs. Hansen and Wang (2021) proposes a soft data augmentation (SODA) method, which adopts bootstrap your own latent (BYOL) (Grill et al., 2020) architecture to maximize representation consistency in latent space. Recently, Wang et al. (2025a) propose a multi-scale temporal contrastive learning (MTCL) method for RL pre-training, which explicitly models multi-scale temporal correlations to capture both motion-aware and static appearance-aware information, yielding more informative representations that enhance policy learning and generalization. Meanwhile, Xiao et al. (2025) have explored enhancing representation robustness through knowledge distillation in heterogeneous or structured feature spaces. Although these methods improve generalization by learning invariant representations under complex environmental variations, they mainly encourage global similarity across observations and lack explicit mechanisms for capturing spatially localized and task-relevant information. As a result, they may struggle to guide the agent's attention toward critical regions in complex visual environments.

Attention and masking-based VRLs aim to address this limitation by explicitly emphasizing salient regions in visual observations and suppressing task-irrelevant information. For instance, the saliency-guided Q-networks (SGQN) (Bertoin et al., 2022) generates saliency-guided masks and incorporates value function consistency to regularize policy learning. Grooten et al. (2024) propose a lightweight masking mechanism (MaDi) driven by reward signals to filter irrelevant distractions. Pore et al. (2024) further leverage the segmentation mask of agent as a supervisory signal to learn decoupled representations of environment and agent through feature separation constraints. More recently, Jiang et al. (2026) combine Jacobian matrix-based transition dynamics modeling with masked contrastive learning (J-Mac) to remove task-irrelevant features from visual state representations, thereby guiding the agent toward task-relevant regions and improving generalization under significant environmental variations. While such VRLs have shown promising results, they typically rely on heuristic mask generation or indirect supervision signals (e.g., reward or saliency), without explicitly modeling the intrinsic relationship between original and augmented observations. Consequently, the learned attention may fail to capture consistent task-relevant regions under visual perturbations.

In contrast, our proposed SSA explicitly models the similarity between original and augmented observations to directly guide attention learning. By leveraging this similarity as a supervisory signal, SSA enables more robust and consistent identification of task-relevant representations, thereby further improving robustness and generalization in unseen environments.

3. Preliminaries

3.1. Problem formulation of VRL

VRL aims to learn decision-making policies from high-dimensional visual observations through continuous interactions between agent and environment. In this paper, we assume that the environment is modeled as an infinite-horizon Markov decision process (MDP), which is defined by a five-tuple $\langle S, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$, where S represents the high-dimensional state space, $\mathcal{A}$ represents the action space, $\mathcal{P} : S \times \mathcal{A} \times S \rightarrow [0, 1]$ is the state transition probability function, $\mathcal{R}(s, a) : S \times \mathcal{A} \rightarrow \mathcal{R}$ denotes the reward function, and $\gamma \in [0, 1]$ is the discount factor, which tradeoff the importance of immediate and future rewards. At any given time step t , the VRL agent, starting in current state $s_t \in S$, selects an action $a_t \in \mathcal{A}$. The environment then transitions to the next state s_{t+1} according to the state transition probability $\mathcal{P}(s_{t+1}|s_t, a_t)$, and the agent receives an immediate reward $r_t = \mathcal{R}(s_t, a_t)$, where the state s_t is typically composed of a stack of k consecutive frames and is defined as $s_t = (o_t, o_{t-1}, \dots, o_{t-k+1})$, in which o_t represents the visual observation at time step t . The objective of VRL is to train the agent to learn an optimal policy $\pi(a_t|s_t) : S \times \mathcal{A} \rightarrow [0, 1]$ to maximize the expected cumulative discounted return $\mathbb{E}_{s_t, a_t \sim \pi} [\sum_{t=0}^{\infty} \gamma^t r_t]$.

3.2. Visual attention mechanisms

When processing complex visual data, visual attention mechanisms (Itti & Koch, 2001) guide the model to allocate more computational resources to crucial parts of the data while ignoring irrelevant parts. Similar to human attention, when viewing an image, we tend to focus on salient regions of interest rather than observing every detail. One common visual attention mechanism is the spatial attention mechanism (Woo et al., 2018), which assigns weights to different regions of the input data based on their importance, thereby enhancing attention to crucial regions. Given an input image $X \in \mathbb{R}^{H \times W \times C}$ with height H , width W , and C channels, the spatial attention mechanism generates a corresponding spatial weight matrix $I \in \mathbb{R}^{H \times W}$, which represents the importance of each pixel, and is expressed as follows:

$$I(i, j) = \sigma(f(X(i, j))), \quad (1)$$

where i and j correspond to the i th row and j th column pixels of input image X , respectively, $f(\cdot)$ represents the convolution or fully connected layer operations, and $\sigma(\cdot)$ is the activation function used to produce normalized weights. These weights are then multiplied element-wise with the corresponding pixels of the input image. Finally, the spatial attention-weighted feature map X' is obtained as follows:

$$X'(i, j, c) = I(i, j) \otimes X(i, j, c), \quad (2)$$

where c denotes the c th channel, and $\otimes$ represents the element-wise product operator. The other attention mechanism, called channel attention (Hu et al., 2018), assigns weights to each channel of the input feature map based on their importance, thus enabling more effective utilization of feature and improving the model's performance. Given an input feature map $X \in \mathbb{R}^{H \times W \times C}$ with C channels, the channel attention mechanism generates a corresponding channel weight vector $V \in \mathbb{R}^C$, indicating the importance of each channel. This can be expressed as:

$$V(c) = \sigma(g(X_c)), \quad (3)$$

where $X_c \in \mathbb{R}^{H \times W}$ represents the feature map of the c th channel, and $g(\cdot)$ refers to the global average pooling (GAP) (Lin et al., 2014) operation. The channel weight vector is then multiplied element-wise with the corresponding channel feature map. Finally, the channel attention-weighted feature map is given by

$$X'(i, j, c) = V(c) \otimes X(i, j, c). \quad (4)$$

In VRL, visual attention mechanisms are primarily used to efficiently process input images or videos, enabling agents to extract crucial visual cues from visual states to optimize decision-making and actions.

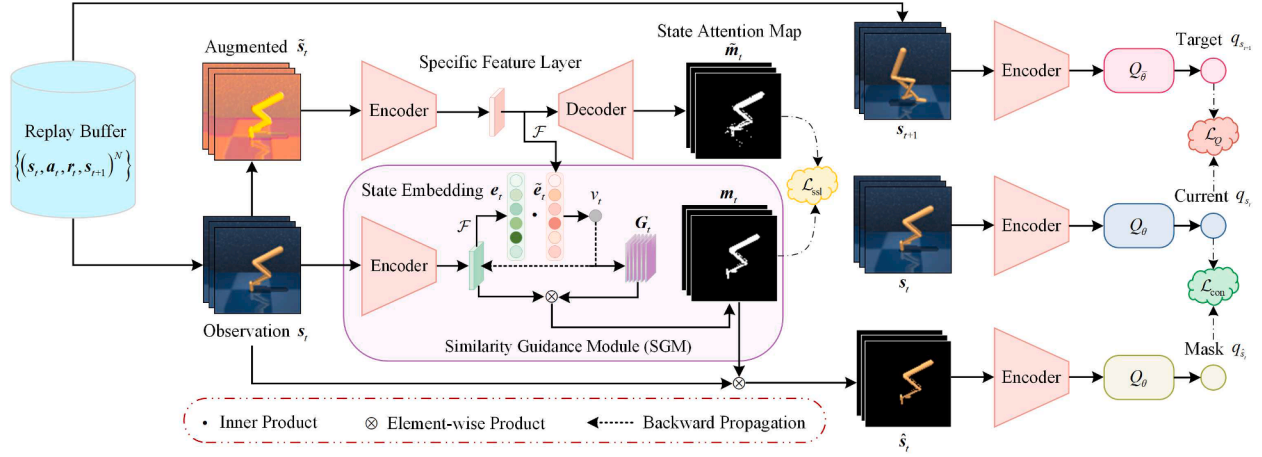


Fig. 1. Overall architecture of similarity-guided state attention for visual reinforcement learning.

Although visual attention mechanisms can effectively ignore irrelevant information and guide agents to focus on task-relevant features, their effectiveness often relies on large amounts of training data. Therefore, combining self-supervised learning can be considered to enhance the agent's learning efficiency.

Self-supervised learning generates supervisory signals directly from the data, without relying on externally labeled data (Bengio et al., 2013; Zhang et al., 2017). Unlike traditional supervised learning, self-supervised learning derives labels from the inherent structure or features within the data and learns effective representations through auxiliary tasks. In VRL, self-supervised learning leverages unlabeled visual data alongside visual attention mechanisms to assist agents in perceiving the environment from visual inputs (e.g., images or videos), enabling more informed decision-making based on learned and extracted features. Especially in the absence of explicit reward signals, self-supervised learning provides additional guidance signals to agents through the structure or patterns inherent in the data, thereby improving the agent's learning efficiency and generalization ability in complex visual environments.

4. Methodology

To better train agents to focus on important regions during decision-making, reduce sensitivity to irrelevant areas, and address robustness and generalization issue in unseen environment, we propose a similarity-guided state attention architecture for VRL in this section, called SSA. First, we outline the overall architecture of SSA. Then, we introduce three main components of SSA: the similarity guidance module based on visual attention mechanism, the similarity measurement in self-supervised learning, and the visual reinforcement learning method with SAC. Finally, we summarize SSA and provide detailed pseudocode.

4.1. Architectural overview

In this paper, we propose a similarity-guided state attention method for VRL and present its architecture, as shown in Fig. 1. The workflow is as follows: firstly, a similarity guidance module (SGM) is constructed. Given the current state and the corresponding action, the current state is augmented, and state representations are extracted. After passing through fully connected layers, low-dimensional state embeddings are obtained via the encoder. The similarity value between the low-dimensional embeddings of current and the augmented states is computed, and this value is backpropagated through gradients to a specific feature layer. This process guides encoder to mine task-relevant representations, thus generating a state attention map. Meanwhile, the attention map of augmented state is produced by decoding its state embedding. Next, a cosine similarity is introduced to measure the global

similarity between attention maps of current and augmented states and combined with BCE as self-supervised learning objective to jointly optimize the encoder and decoder, thereby refining state attention maps. Finally, the current state is fused with its refined state attention map to obtain a mask, which is used for RL to estimate Q-value. Simultaneously, Q-value estimation is performed using current state. A consistency term is used to compute the degree of consistency between the two Q-values, regularizing the value function learning objective and encouraging the value function to prioritize significant crucial pixels, thereby obtaining the optimal policy.

4.2. Similarity guidance module based on visual attention mechanism

The similarity guidance module (SGM) draws inspiration from visual attention mechanisms (Itti & Koch, 2001) and Grad CAM (Selvaraju et al., 2020). Its purpose is to guide the agent's self-awareness of the importance of the region in which it is located by computing the similarity between the current and augmented state representations. This process helps reduce the interference from task-irrelevant distractions and effectively distinguishes between important and unimportant pixels in the state embeddings. We begin by sampling experience $(s_t, a_t, r_t, s_{t+1}) \sim \mathbb{B}$ from replay buffer $\mathcal{D}$, where $\mathbb{B} \in \mathcal{D}$ denotes a mini-batch of size B . A random convolution is applied to current visual state observation s_t to generate augmented visual state observation $\tilde{s}_t$. Both s_t and $\tilde{s}_t$ are encoded by the encoder f_ρ with parameter ρ , and then passed through the fully connected layer F to obtain d -dimensional state embeddings $e_t = F(f_\rho(s_t), a_t)$ and $\tilde{e}_t = F(f_\rho(\tilde{s}_t), a_t)$. The similarity v_t between e_t and $\tilde{e}_t$ is computed via the inner product as follows:

$$v_t = e_t^\top \tilde{e}_t. \quad (5)$$

The inner product serves as a simple and efficient similarity measurement, where larger values indicate higher similarity. Compared with raw pixel-based similarity, which is sensitive to low-level variations such as lighting and texture changes, the state embedding-based similarity is computed over learned representations that contain higher-level semantic and task-relevant information. This reduces sensitivity to visual perturbations and enables more stable similarity estimation, thereby guiding the encoder to focus on invariant task-relevant representations rather than superficial pixel-level details.

Next, v_t is backpropagated to a specific feature layer $F \in \mathbb{R}^{W \times H \times C}$ within f_ρ , and its feature gradient matrix G_t (i.e., $G_t = \frac{\partial v_t}{\partial F}$) is computed to select features that are more similar to augmented visual state observation $\tilde{s}_t$. During backpropagation, G_t enables that the target feature positions with larger gradients are prioritized for adjustment, thereby increasing the similarity between e_t and $\tilde{e}_t$. Subsequently, the importance of each channel in the specific feature layer is measured using the

mean of G_t . Specifically, a channel attention mechanism is employed to measure the importance of each channel, that is, applying GAP to G_t , which yields the channel weight:

$$\omega_t^c = \frac{1}{W \times H} \sum_{i=1}^W \sum_{j=1}^H \frac{\partial v_t}{\partial F_{ij}^{tc}}, \quad (6)$$

where $F^t \in \mathbb{R}^{W \times H \times C}$ represents the features of s_t at the specified feature layer F , W and H denote the width and height of feature, respectively, F^{tc} represents the c th channel feature of F^t , F_{ij}^{tc} refers to the feature pixel of the i th row and j th column in F^{tc} , ω_t^c represents the weight of G_t for the c th channel feature corresponding to s_t . Finally, the obtained channel weight ω_t^c is redistributed to F^t , guiding the module to focus on crucial visual regions that exhibit high similarity, thereby generating the state attention map m_t for s_t :

$$m_t = \text{ReLU}\left(\sum_{c=1}^C \omega_t^c F^{tc}\right), \quad (7)$$

where $\text{ReLU}(\cdot)$ denotes the activation function. In addition, the augmented visual state observation $\tilde{s}_t$ and current action a_t are fed into encoder f_ρ and decoder M_ρ to obtain the state attention map $\tilde{m}_t$ (i.e., $\tilde{m}_t = M_\rho(f_\rho(\tilde{s}_t), a_t)$). The $\tilde{m}_t$ serves as a supervisory signal, providing guidance for generating m_t of s_t . The process constitutes self-supervised learning.

4.3. Similarity measurement in self-supervised learning

To more accurately and comprehensively capture the crucial pixel regions that influence the agent's decision-making and further refine the state attention maps, this section introduces a cosine similarity term in the self-supervised learning objective to measure the global similarity between the state attention maps m_t and $\tilde{m}_t$. Specifically, we use the projection operation $p(\cdot)$ to obtain representation vectors $z_t = p(m_t)$ and $\tilde{z}_t = p(\tilde{m}_t)$, which preserve the global information of state attention maps while providing compact task-relevant representations. The global similarity of state attention maps is then measured using cosine similarity between z_t and $\tilde{z}_t$:

$$\mathcal{L}_{\cos}(\rho) = \frac{1}{B} \sum_{b=1}^B \left(1 - \frac{z_t^b \cdot \tilde{z}_t^b}{\|z_t^b\| \|\tilde{z}_t^b\|} \right). \quad (8)$$

This cosine similarity measures the directional alignment between state attention maps, encouraging the f_ρ and M_ρ to maintain consistent focus on globally important regions across different augmented views of the same state. In addition, to enhance the precision of pixel-level saliency in the state attention maps, the BCE between m_t and $\tilde{m}_t$ is computed as:

$$\mathcal{L}_{\text{BCE}}(\rho) = \frac{1}{B} \sum_{b=1}^B \left[-\tilde{m}_t^b \log(m_t^b) - (1 - \tilde{m}_t^b) \log(1 - m_t^b) \right]. \quad (9)$$

The BCE loss focuses on pixel-level accuracy and complements the global similarity, allowing the model to capture both fine-grained local details and global attention consistencies.

Thus, the overall self-supervised learning objective is:

$$\mathcal{L}_{\text{ssl}}(\rho) = \mathcal{L}_{\text{BCE}}(\rho) + \lambda \mathcal{L}_{\cos}(\rho), \quad (10)$$

where λ is the weight coefficient.

4.4. Visual reinforcement learning with SAC

In Sections 4.2 and 4.3, the similarity-guided state attention is formulated as a general self-supervised learning task. The current state attention map m_t and augmented state attention map $\tilde{m}_t$ are refined through the self-supervised learning objective. Furthermore, m_t is combined with the original state observation s_t to create a mask $\hat{s}_t$ (i.e., $\hat{s}_t = m_t \otimes s_t$),

which provides crucial visual information for downstream RL task, enabling the agent to learn optimal policies. In this paper, we adopt soft actor-critic (SAC) (Haarnoja et al., 2018) as the foundational RL method. SAC, an off-policy Actor-Critic method, proposes maximizing reward and policy entropy, which fosters exploration, improves robustness, and encourages policy diversity. The method effectively preventing premature convergence to local optima, making it well-suited for continuous control tasks. We combine the aforementioned self-supervised learning with SAC to instantiate a practical VRL method, referred to as SSA. Next, this section provides a detailed explanation of SSA implementation.

SSA uses neural network parameter ρ to parameterize encoder f_ρ and decoder M_ρ in self-supervised learning. It also parameterizes two Critic networks $Q_{\theta_1}, Q_{\theta_2}$, one Actor network π_ψ , using parameters θ_1, θ_2 , and ψ , respectively, in SAC. Meanwhile, parameters $\bar{\theta}_1$ and $\bar{\theta}_2$ are used to parameterize two targets $Q_{\bar{\theta}_1}$ and $Q_{\bar{\theta}_2}$ for stable training. Experience $(s_t, a_t, r_t, s_{t+1}) \sim \mathbb{B}$ is sampled from replay buffer $\mathcal{D}$, and Q-function is approximated by minimizing the soft Bellman error:

$$\mathcal{L}_Q(\theta_h) = \mathbb{E}_{(s_t, a_t, r_t, s_{t+1}) \sim \mathbb{B}} \left[Q_{\theta_h}(f_\rho(s_t), a_t) - B^\pi Q(f_\rho(s_t), a_t) \right]^2, \quad (11)$$

where $B^\pi Q(f_\rho(s_t), a_t) = r_t + \gamma \mathbb{E}_{s_{t+1} \sim P(\cdot | s_t, a_t), a_{t+1} \sim \pi} [Y]$, $Y = Q_{\bar{\theta}_h}(f_\rho(s_{t+1}), a_{t+1}) - \alpha \log \pi_\psi(a_{t+1} | f_\rho(s_{t+1}))$, B^π is a Bellman operator, θ_h represents the network parameter of Q_{θ_h} , $h = 1, 2$, $\bar{\theta}_h$ is the network parameter of target $Q_{\bar{\theta}_h}$ corresponding to Q_{θ_h} , which is updated by the exponential moving average (EMA):

$$\bar{\theta}_h \leftarrow \tau \theta_h + (1 - \tau) \bar{\theta}_h, \quad (12)$$

where τ is the updating rate of target network. In addition, according to the setting in Bertoin et al. (2022), the consistency measurement is performed between Q-value estimations of mask $\hat{s}_t$ and original state s_t to encourage the Q-function to maintain consistent Q-value prediction even when specific perturbations are applied to the input observation:

$$\mathcal{L}_{\text{con}}(\theta_h) = \mathbb{E}_{(s_t, a_t, r_t, s_{t+1}) \sim \mathbb{B}} \left[Q_{\theta_h}(f_\rho(s_t), a_t) - Q_{\theta_h}(f_\rho(\hat{s}_t), a_t) \right]^2. \quad (13)$$

Therefore, the objective function of Critic network is jointly implemented by Eqs. (11) and (13):

$$\mathcal{L}_{\text{critic}}(\theta_h) = \mathcal{L}_Q(\theta_h) + \mathcal{L}_{\text{con}}(\theta_h). \quad (14)$$

The objective function of Actor network consists of maximizing the Q-function value and policy entropy regularization term:

$$\mathcal{L}_{\text{actor}}(\psi) = -\mathbb{E}_{a_t \sim \pi} [Q_{\theta_h}(f_\rho(s_t), a_t) - \alpha \log \pi_\psi(a_t | f_\rho(s_t))], \quad (15)$$

where α is the temperature coefficient, which determines the relative importance of the entropy term with respect to the reward.

Algorithm 1 presents the pseudocode for training processes of SSA. Initially, the agent processes the current state, performs an action, and receives the corresponding reward and next state from the environment. This process iterates, generating a set of states, actions, and rewards, which are stored as experience data in a replay buffer. During each update, experience is sampled from the buffer for updating parameters of the self-supervised learning, Critic, Actor, and target networks. Once the training reaches the predefined number of steps, the agent is evaluated in a test environment to assess the robustness and generalization ability of learned policy to unseen environments. The training phase returns the updated parameters for the self-supervised learning, Critic, Actor, and target networks.

5. Experiments

In this section, experiments are conducted on five visual continuous control tasks from DMControl-GB (Hansen & Wang, 2021) to evaluate the overall robustness and generalization capability of SSA. The DMControl-GB is derived from the DeepMind Control Suite (DMControl) (Tassa et al., 2018), with modifications to the background complexity, making it a more challenging benchmark for generalization.

Algorithm 1 SSA training.

Input: Batch size B , learning rates φ_{actor} , φ_{critic} , and φ_{ssl} , discount factor γ , temperature coefficient α , updating rate of target network τ , and weight coefficient λ .

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1: Initialize: Replay buffer  $\mathcal{D}$ , Encoder parameter  $\rho$ , Actor parameter  $\psi$ , Critic parameter  $\theta$ , and Target parameter  $\bar{\theta}$ .
2: for each interaction step do
3:   for each time step do
4:     Collect transition  $(s_t, \mathbf{a}_t, r_t, s_{t+1})$ ;
5:     Add transition to replay buffer  $\mathcal{D} \leftarrow \mathcal{D} \cup \{(s_t, \mathbf{a}_t, r_t, s_{t+1})\}$ ;
6:   end for
7:   for each update step do
8:     Sample mini-batch  $\mathbb{B}$  from  $\mathcal{D}$ ;
9:      $(s_t, \mathbf{a}_t, r_t, s_{t+1}) \sim \mathbb{B}$ ;
10:    // Self-supervised learning update
11:    Update  $\rho \leftarrow \rho - \varphi_{\text{ssl}} \nabla_{\rho} \mathcal{L}_{\text{ssl}}(\rho)$  via Eq. (10);
12:    // Actor update
13:    Update  $\psi \leftarrow \psi - \varphi_{\text{actor}} \nabla_{\psi} \mathcal{L}_{\text{actor}}(\psi)$  via Eq. (15);
14:    // Critic update
15:    Update  $\theta_h \leftarrow \theta_h - \varphi_{\text{critic}} \nabla_{\theta_h} \mathcal{L}_{\text{critic}}(\theta_h)$  for  $h = \{1, 2\}$  with Eq. (14);
16:    // Target Network update
17:    Update  $\bar{\theta}_1$  and  $\bar{\theta}_2$  by Eq. (12);
18:  end for
19: end for
Output:  $\rho$ ,  $\psi$ ,  $\theta$ , and  $\bar{\theta}$ .

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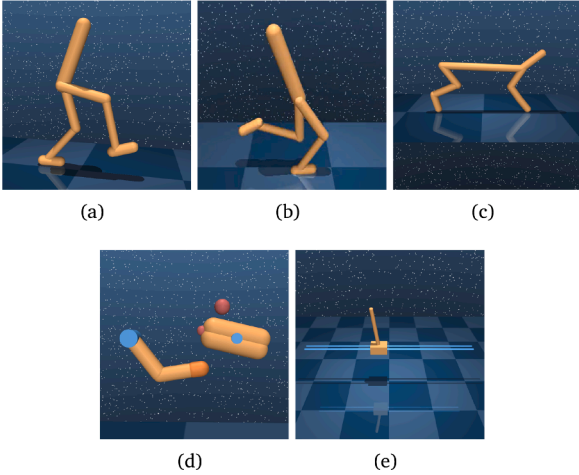


Fig. 2. Pictorial illustration of five control tasks. (a) walker walk. (b) walker stand. (c) cheetah run. (d) finger spin. (e) cartpole swingup.

5.1. Experiment setup

This section provides a description of experimental setup, including the experimental environments, baselines, hyperparameter setting, implementation detail and evaluation protocol.

1) *Environments and Baselines:* We conducted experiments on five control tasks, including Walker Walk, Walker Stand, Cheetah Run, Finger Spin, and Cartpole Swingup, as shown in Fig. 2. During training, the input tasks are provided by the DMControl environment, while during testing, the unseen environments are drawn from the extended DMControl-GB, which is based on DMControl. DMControl-GB introduces variations in the environment backgrounds distribution, including random color backgrounds and natural video backgrounds. To ensure fair comparison with existing VRL methods, SSA is tested in the common natural video background environments provided by DMControl-GB, evaluating its robustness and generalization ability in unseen environments.

Table 1

Specification of state, action, reward, and observation spaces for VRL tasks.

Task	dim($\mathcal{S}$)	dim($\mathcal{A}$)	$\mathcal{R}(s, a)$	dim($\mathcal{O}$)
Walker Walk	18	6	[0, 1]	$84 \times 84 \times 3$
Walker Stand	18	6	[0, 1]	$84 \times 84 \times 3$
Cheetah Run	18	6	[0, 1]	$84 \times 84 \times 3$
Finger Spin	6	2	[0, 1]	$84 \times 84 \times 3$
Cartpole Swingup	4	1	[0, 1]	$84 \times 84 \times 3$

Table 2

Hyperparameter setting of SSA.

Hyperparameter	Value
Observation rendering	(84, 84, 3)
Frame stack	3
Random seed	5
Replay buffer $\mathcal{D}$	500,000
Max time step	500,000
Discount factor γ	0.99
Episode length	1000
Augmentation	Random convolution
Evaluation frequency	10,000
Batch size B	128
Target networks update frequency	2
Temperature coefficient α	0.1
Updating rate of target network τ	0.01
Learning rates	
φ_{actor}	1e-3
φ_{critic}	1e-3
φ_{ssl}	5e-4
Action repeat	
Finger spin	2
Cartpole swingup	8
Otherwise	4
Weight coefficient λ	0.8

The natural video backgrounds are randomly sampled from the Places dataset as representatives of the unseen environments, including two difficulty levels: video_easy, which introduces modest background variation, and video_hard, which contains stronger environment distractors such as cluttered or highly dynamic scenes. This setup allows us to assess SSA's robustness and generalization compared with VRL baselines in both standard and distractor-heavy environments. For intuitive understanding of the reinforcement learning setup, Table 1 summarizes the state, action, reward, and observation specifications for all tasks. Observations consist of RGB images of size $84 \times 84 \times 3$, and rewards are defined as scalars within [0, 1].

SAC (Haarnoja et al., 2018) serves as the backbone RL algorithm in this study, and SSA is compared with several state-of-the-art VRL baselines, including RAD (Laskin et al., 2020a), DrQ (Yarats et al., 2021b), CURL (Laskin et al., 2020b), SODA (Hansen & Wang, 2021), SGQN (Bertoin et al., 2022), MaDi (Grooten et al., 2024), J-Mac (Jiang et al., 2026), and DGA2 (Lee & Hwang, 2025). All methods are implemented using PyTorch 1.10 on an NVIDIA GeForce RTX 2080 Ti GPU and an Intel i7-12700K CPU to ensure fair comparison.

2) *Hyperparameter Setting:* Table 2 summarizes the hyperparameter setting required for training SSA, where the action repeat parameters are consistent with the setting provided in Hansen and Wang (2021).

3) *Implementation Detail and Evaluation Protocol:* SSA is a general framework that can be integrated with other standard RL method. SAC is used as the backbone method to implement SSA in this paper. In self-supervised learning network, the constructed SGM encoder f_{ρ} consists of 11 convolutional layers, each with 32 filters of 3×3 kernels and ReLU activation units. The first convolutional layer has a stride of 2, while the others have a stride of 1, with zero padding. The input of f_{ρ} consists of three stacked frames of size $84 \times 84 \times 3$, and the output feature size is $32 \times 21 \times 21$. This feature is flattened into a 1×100 vector via an additional fully connected layer, and the gradient feature obtained through

Table 3

Evaluation return of SSA and VRL baselines in the last 10% time steps. Each evaluation return report reported the standard deviation of 5 random seeds. The **bold** and underlined values represent the **optimal** and **suboptimal** values, respectively. Δ represent the difference from the suboptimal value.

DMControl-GB		Evaluation Return										
		SAC	RAD	DrQ	CURL	SODA	SGQN	MaDi	J-Mac	DGA2	SSA	Δ
Video_Easy	Walker Walk	338 $\pm$ 71	654 $\pm$ 59	591 $\pm$ 84	648 $\pm$ 47	482 $\pm$ 86	735 $\pm$ 332	642 $\pm$ 77	591 $\pm$ 83	518 $\pm$ 60	804 $\pm$ 80	+9.39%
	Walker Stand	481 $\pm$ 49	881 $\pm$ 59	853 $\pm$ 76	<u>891 $\pm$ 63</u>	802 $\pm$ 99	307 $\pm$ 316	798 $\pm$ 231	818 $\pm$ 231	772 $\pm$ 67	949 $\pm$ 25	+6.51%
	Cheetah Run	182 $\pm$ 38	161 $\pm$ 27	180 $\pm$ 23	<u>184 $\pm$ 32</u>	165 $\pm$ 45	294 $\pm$ 15	186 $\pm$ 18	149 $\pm$ 46	<u>212 $\pm$ 20</u>	<u>212 $\pm$ 23</u>	-27.89%
	Finger Spin	332 $\pm$ 118	560 $\pm$ 75	507 $\pm$ 55	490 $\pm$ 30	359 $\pm$ 99	565 $\pm$ 9	346 $\pm$ 23	754 $\pm$ 47	<u>546 $\pm$ 18</u>	<u>629 $\pm$ 72</u>	-16.58%
	Cartpole Swingup	460 $\pm$ 62	378 $\pm$ 85	474 $\pm$ 72	458 $\pm$ 74	375 $\pm$ 65	507 $\pm$ 343	443 $\pm$ 69	472 $\pm$ 97	<u>704 $\pm$ 55</u>	711 $\pm$ 102	+0.99%
	Average	358.6	526.8	521.0	534.2	436.6	481.6	483.0	<u>556.8</u>	550.4	661.0	+18.71%
Video_Hard	Walker Walk	154 $\pm$ 64	71 $\pm$ 21	115 $\pm$ 41	58 $\pm$ 14	129 $\pm$ 85	629 $\pm$ 116	-	292 $\pm$ 50	362 $\pm$ 68	<u>586 $\pm$ 26</u>	-6.84%
	Walker Stand	271 $\pm$ 78	226 $\pm$ 37	297 $\pm$ 53	226 $\pm$ 44	258 $\pm$ 62	154 $\pm$ 26	-	<u>730 $\pm$ 102</u>	689 $\pm$ 63	799 $\pm$ 34	+9.45%
	Cheetah Run	68 $\pm$ 16	48 $\pm$ 22	48 $\pm$ 13	18 $\pm$ 17	28 $\pm$ 11	159 $\pm$ 48	-	<u>103 $\pm$ 25</u>	153 $\pm$ 23	<u>155 $\pm$ 12</u>	-2.52%
	Finger Spin	46 $\pm$ 31	73 $\pm$ 50	60 $\pm$ 21	47 $\pm$ 25	27 $\pm$ 15	485 $\pm$ 106	-	400 $\pm$ 48	431 $\pm$ 62	492 $\pm$ 23	+1.44%
	Cartpole Swingup	158 $\pm$ 16	132 $\pm$ 23	113 $\pm$ 21	120 $\pm$ 15	128 $\pm$ 26	208 $\pm$ 155	-	214 $\pm$ 46	<u>499 $\pm$ 50</u>	533 $\pm$ 28	+6.81%
	Average	139.4	110.0	126.6	93.8	114.0	327.0	-	347.8	<u>426.8</u>	513.0	+20.19%

backpropagation matches the output dimension of the last feature layer in f_ρ , which is $32 \times 21 \times 21$. The augmented stacked frames share the encoder f_ρ and F layer. The decoder M_ρ consists of 3 convolutional layers and upsampling, with the first convolutional layer using 128 filters of 3×3 kernels and ReLU activation units, the second convolutional layer using 64 filters of 3×3 kernels, and the last convolutional layer using 9 filters of 3×3 kernels, all with padding of 1. The resulting state attention map and mask have the same dimensions as the input frames. In SAC, both the Actor and Critic networks are composed of an encoder and a multilayer perceptron (MLP), where the encoder structure is the same as the one described above. The MLP consists of 2 ReLU activation units and 3 linear layers with 1024 hidden units.

In addition, We adopt a publicly available evaluation return as the evaluation metric to measure the overall performance of each method in this paper, which represents the cumulative reward of the agent in a single episode consisting of 1000 time steps. During the experiment, the agent is trained for 500,000 time steps across all control tasks, evaluated the VRL model every 10,000 time steps, and recorded the reward. For each evaluation, a deterministic policy is sampled on the agent, and the average action distribution is adopted. The average return over every 10,000 time steps is computed across all random seeds to evaluate the overall performance of agent.

5.2. Experimental results and analysis

We conduct extensive evaluations of SSA on the DMControl-GB benchmark, focusing on robustness and generalization performance, sample efficiency, and computational cost. The analysis is organized into evaluations on video_easy and video_hard environments, visualization analysis, and computational complexity analysis.

5.2.1. Evaluation on video_easy and video_hard environments

Table 3, Figs. 3, and 4 present the evaluation performance of SSA and all baselines on both video_easy and video_hard environments. The reported results correspond to the average evaluation returns over the final 10% of time steps across five random seeds, and the shaded regions in the figures denote the standard deviation. By jointly analyzing the quantitative results and evaluation return curves under different levels of visual disturbance, the following observations can be drawn.

1) From the overall evaluation perspective, SSA achieves consistently strong performance across both environments. On the video_easy environment, SSA obtains the highest average evaluation return of 661.0, exceeding the suboptimal method J-Mac by 18.71%. This advantage is particularly evident in Walker tasks, where SSA reaches

804 in Walker Walk and 949 in Walker Stand, significantly outperforming all baselines. On the more challenging video_hard environment, SSA maintains its superiority with an average evaluation return of 513.0, surpassing DGA2 by 20.19%. The consistent improvement under increasing visual disturbance indicates that SSA effectively enhances robustness and generalization capability under complex environment variations.

2) From the task-level perspective, SSA demonstrates consistently stable performance across diverse control tasks under varying levels of visual disturbance. On locomotion tasks such as Walker Walk and Walker Stand, SSA achieves high evaluation returns on both video_easy and video_hard environments, reaching 804 and 949 on video_easy, and 586 and 799 on video_hard, respectively. In comparison, methods such as SGQN exhibit strong performance on specific tasks, particularly achieving 294 in Cheetah Run on video_easy and 629 in Walker Walk on video_hard. However, these advantages are not consistently maintained and are accompanied by substantial performance variance, such as 735 in Walker Walk and 307 in Walker Stand on video_easy. This phenomenon can be attributed to the sensitivity of SGQN to training conditions and its strong dependence on the hyperparameter configuration of auxiliary supervision, which may induce overfitting and gradient imbalance, thereby limiting robustness and generalization performance. J-Mac achieves strong performance on tasks such as Finger Spin, reaching 754 on video_easy, which is related to its Jacobian-based representation learning that emphasizes local feature sensitivity. However, its performance on locomotion tasks requiring global dynamics understanding remains limited, as reflected by an evaluation return of 591 in Walker Walk. DGA2 presents relatively balanced performance across tasks, achieving 704 in Cartpole Swingup on video_easy and 499 on video_hard, yet its overall performance remains consistently below that of SSA. Overall, SSA maintains moderate variance and smoother learning curves across tasks, indicating more stable learning and improved robustness to environmental variations.

3) From the perspective of sample efficiency, SSA demonstrates competitive convergence performance. As illustrated in Figs. 3 and 4, on the video_easy environment, SSA reaches high evaluation returns at earlier stages, achieving evaluation returns of 800 and 900 in Walker Walk and Walker Stand at approximately 280,000 and 60,000 time steps, respectively, and subsequently converges in a stable manner. Under the same 100,000 time steps, SSA consistently attains higher evaluation returns, reaching about 600 in Walker Walk and 900 in Walker Stand, whereas other methods remain below these performance levels. In comparison, J-Mac requires substantially more time steps to reach similar performance levels, while SGQN, although exhibiting rapid initial improvement, fails to maintain stable

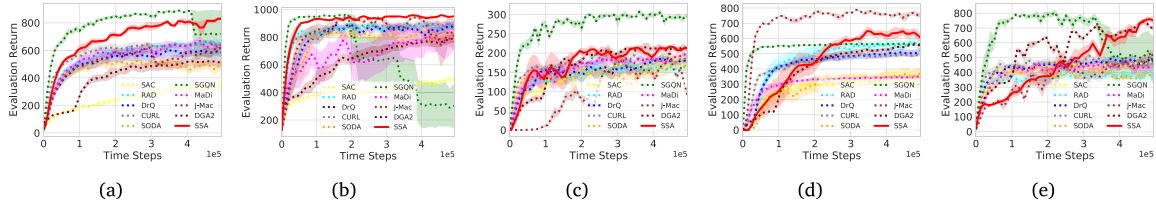


Fig. 3. Evaluation return curves of SSA against VRL baselines on the video_easy environment across five control tasks. (a) walker walk. (b) walker stand. (c) cheetah run. (d) finger spin. (e) cartpole swingup.

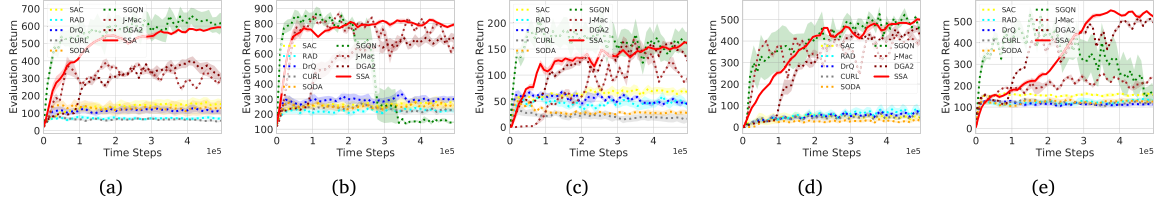


Fig. 4. Evaluation return curves of SSA against VRL baselines on the video_hard environment with five control tasks. (a) walker walk. (b) walker stand. (c) cheetah run. (d) finger spin. (e) cartpole swingup.

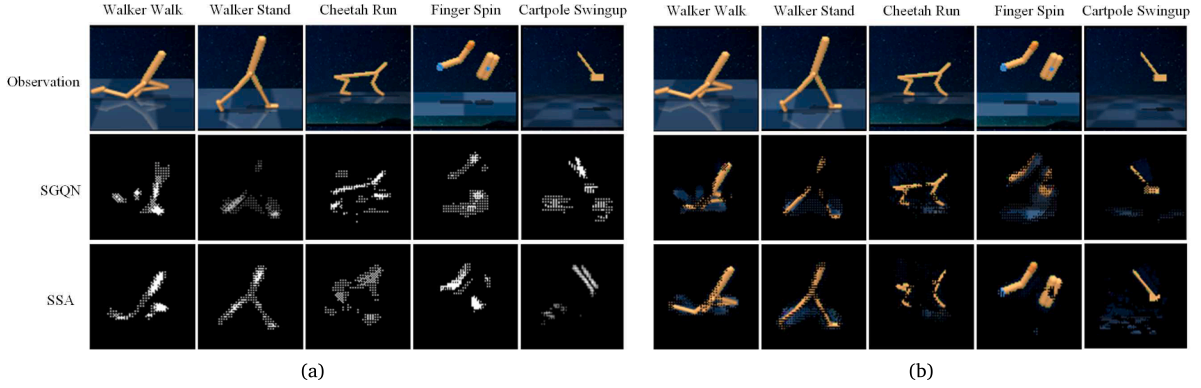


Fig. 5. Visualization results of SSA and SGQN on the video_easy environment across five control tasks. (a) state attention maps highlighting task-relevant regions. (b) corresponding mask maps used for policy learning.

convergence. A similar trend is observed on the video_hard environment, where SSA maintains steady performance improvement despite stronger visual disturbances. The learning curves show a consistent upward trend with reduced fluctuations, indicating that SSA effectively utilizes training samples while preserving stable learning.

- 4) Overall, the results across both environments demonstrate that SSA achieves a favorable trade-off among robustness and generalization performance. Compared with existing VRL methods, SSA not only improves evaluation returns but also exhibits reduced sensitivity to visual perturbations and more consistent performance across tasks. This suggests that the proposed similarity-guided representation learning effectively captures task-relevant features and supports robust policy learning in complex visual environments.

5.2.2. Visualization analysis

To qualitatively evaluate whether the learned representations can accurately capture task-relevant regions for policy learning, we visualize the state attention maps and corresponding mask maps of SSA and SGQN across five control tasks on the video_easy environment, as shown in Fig. 5. From the state attention maps, as shown in Fig. 5(a), SSA consistently focuses on task-relevant regions that align well with the agent dynamics. For locomotion tasks such as Walker Walk and Walker Stand, the attention concentrates on the articulated body and joints, while in Cheetah Run, it captures the overall body structure under rapid mo-

tion. For manipulation tasks such as Finger Spin and Cartpole Swingup, SSA accurately highlights the interacting objects and their local regions. By comparison, SGQN also identifies relevant regions but often produces more scattered attention across tasks. These differences are further reflected in the mask maps, as shown in Fig. 5(b). SSA generates more spatially coherent and structurally consistent masks that closely match the geometry of task-relevant objects, effectively suppressing irrelevant background information. In contrast, the masks produced by SGQN, while still informative, tend to include more dispersed and less task-focused regions. Overall, the results indicate that SSA is able to more accurately and consistently capture task-relevant regions, leading to more informative visual representations for policy learning.

5.2.3. Computational complexity analysis

To evaluate the computational efficiency of SSA, Table 4 summarizes the training time and testing time for all compared VRL methods under the same experimental settings.

- 1) From Table 4, SSA requires 20.8 h to train a single task. This is higher than lightweight baselines such as SAC, RAD, and DrQ, but remains competitive among methods that incorporate auxiliary learning objectives or attention mechanisms. In particular, SSA is more efficient than SGQN, J-Mac, and DGA2, which incur substantially higher training costs due to more complex architectures or additional optimization procedures. This indicates that SSA introduces a moderate

Table 4
Training time and testing time for each evaluated VRL method.

Method	SAC	RAD	DrQ	CURL	SODA	SGQN	MaDi	J-Mac	DGA2	SSA
Training time (h/task)	5.0	7.8	7.6	11.7	12.4	22.9	19.9	25.5	33.0	20.8
Testing time (s/episode)	1.9	2.2	2.3	3.5	3.6	3.7	2.6	2.9	4.1	2.6

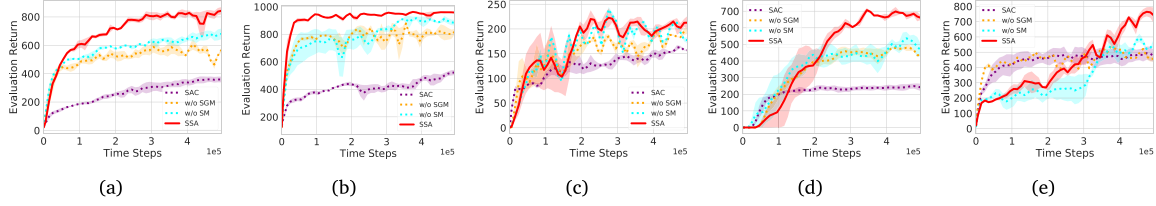


Fig. 6. Ablation study. Comparison of performance curves with incremental addition of SGM and cosine similarity-based alignment of state attention maps $\mathcal{L}_{\cos}$ on DMControl-GB in the same seed setting. (a) walker walk. (b) walker stand. (c) cheetah run. (d) finger spin. (e) cartpole swingup.

Table 5
Ablation study of proposed SSA.

DMControl-GB	Evaluation Return			
	SAC	w/o SGM	w/o SM	SSA
Walker Walk	338 ± 71	552 ± 63	683 ± 52	804 ± 80
Walker Stand	481 ± 49	798 ± 74	884 ± 36	949 ± 25
Cheetah Run	182 ± 38	185 ± 15	191 ± 21	212 ± 23
Finger Spin	332 ± 118	467 ± 31	519 ± 67	629 ± 72
Cartpole Swingup	460 ± 62	471 ± 41	526 ± 72	711 ± 102
Average	358.6	494.6	560.6	661.0

increase in training cost while maintaining a reasonable level of efficiency within its category.

- In terms of inference, SSA requires 2.6 s per episode, which is comparable to most baselines, indicating that SSA does not introduce noticeable deployment overhead. The additional computational cost of SSA mainly arises during training due to the similarity-guided attention mechanism, which introduces extra forward and backward computations through additional gradient propagation paths in the SGM. Compared with representation learning methods such as CURL and SODA, as well as attention-based methods such as SGQN and DGA2, SSA maintains a favorable trade-off between efficient inference process and computational efficiency while achieves improved robustness and generalization performance.

5.3. Ablation study

To investigate the individual contributions of SGM and cosine similarity-based alignment of state attention maps $\mathcal{L}_{\cos}$, we conduct ablation experiments on five control tasks from DMControl-GB, with results reported in Fig. 6 and Table 5. In addition to the baseline SAC and SSA, we consider two intermediate variants: w/o SGM, which removes the similarity guidance module, and w/o SM, which removes the cosine similarity-based alignment of state attention maps $\mathcal{L}_{\cos}$. This design enables a clear analysis of the role of each component.

As shown in Fig. 6 and Table 5, the w/o SGM variant, which applies similarity supervision without the SGM attention mechanism, achieves only limited improvement over SAC across most tasks. This indicates that the cosine similarity-based alignment of state attention maps $\mathcal{L}_{\cos}$ alone is insufficient to effectively guide representation learning without explicit attention modeling. In contrast, the w/o SM variant, which introduces SGM without $\mathcal{L}_{\cos}$, yields substantial performance gains over both SAC and w/o SGM. This demonstrates that SGM plays a dominant role by directing the model to focus on task-relevant regions while sup-

pressing irrelevant information, thereby significantly improving representation quality. Furthermore, the full SSA consistently achieves the best robustness and generalization performance across all tasks, indicating that the cosine similarity-based alignment of state attention maps $\mathcal{L}_{\cos}$ provides complementary benefits by enforcing global similarity between attention maps of original and augmented states, which further refines the representations learned by SGM.

Overall, the ablation results show that SGM serves as the primary contributor to performance improvements, while the cosine similarity-based alignment of state attention maps $\mathcal{L}_{\cos}$ plays a complementary role in enhancing robustness and generalization. Their integration leads to the most effective and stable performance.

6. Conclusion

In this paper, we propose a similarity guided state attention for VRL, called SSA. In SSA, (i) a visual attention mechanism-based SGM is designed to calculate the similarity between state embeddings of original and augmented observations, guiding the encoder to extract task-relevant representations and producing a state attention map of original observation that effectively suppresses redundant information and irrelevant factors; (ii) we introduce cosine similarity to measure the global similarity between attention maps of original and augmented states, and jointly use it with BCE as self-supervised learning objective to refine state attention maps. The refined state attention map is further fused with original observation as a mask, providing more informative pixel-level support for downstream RL, thus improving the agent's robustness and generalization performance in unseen environments. To validate the advantages of SSA in robustness and generalization performance, we conduct experiments on DMControl-GB. The results show that, compared with representative VRL baselines, SSA exhibits higher robustness and stronger generalization ability. Compared with the basic SAC, SSA improves the overall average evaluation return by 84.33% and 268.01%, respectively. Ablation experiments further verified the crucial role of SGM and cosine similarity-based alignment of state attention maps in improving robustness and generalization performance. These results indicate that SSA learns decision-aware representations through similarity-guided state attention, enabling more robust and task-relevant representation extraction under visual perturbations. Nevertheless, further strengthening the tight integration between representation learning and policy optimization, as well as improving robustness and generalization under complex environment variations, remains an important direction. In addition, extending SSA to leverage richer multimodal information, such as visual-language signals, and to more realistic scenarios including robotic control, represents a promising avenue for future research.

CRedit authorship contribution statement

Yinfeng Zeng: Writing – original draft, Software, Methodology, Investigation, Conceptualization; **Xuesong Wang:** Writing – review & editing, Supervision, Resources; **Yuhu Cheng:** Writing – review & editing, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is funded by the National Natural Science Foundation of China under Grant 62373364, and Grant 62573416.

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